1.6 Adding Your Own Functions

1.6.1 Defining Functions

In this chapter, we have seen many examples of functions that are built into *Mathematica*. In this section, we discuss how you can add your own simple functions to *Mathematica*. Chapter 2 will describe in much greater detail the mechanisms for adding functions to *Mathematica*.

As a first example, consider adding a function called `f` which squares its argument. The *Mathematica* command to define this function is `f[x_] := x^2`. The `_` (referred to as "blank") on the left-hand side is very important; exactly what it means will be explained in Section 2. For now, just remember to put a `_` on the left-hand side, but not the right-hand side, of your definition.

This defines the function `f`. Notice the `_` on the left-hand side.

```
In[1]:= f[x_] := x^2
```

`f` squares its argument.

```
In[2]:= f[a+1]

                 2
Out[2]= (1 + a)
```

The argument can be a number.

```
In[3]:= f[4]

Out[3]= 16
```

Or it can be a more complicated expression.

```
In[4]:= f[3x + x^2]

                   2 2
Out[4]= (3 x + x )
```

You can use `f` in a calculation.

```
In[5]:= Expand[f[(x+1+y)]]

                     2                    2
Out[5]= 1 + 2 x + x  + 2 y + 2 x y + y
```

This shows the definition you made for `f`. What `/:` means will be discussed in Chapter 2.

```
In[6]:= ?f

f

                 2
f/: f[x_] := x
```

`f[x_] := x^2`	define the function `f`
`?f`	show the definition of *f*
`Clear[f]`	clear all definitions for *f*

Defining a function in *Mathematica*.

The names like `f` that you use for functions in *Mathematica* are just symbols. As a result, you should make sure to avoid using names that begin with capital letters, to prevent confusion with built-in *Mathematica* functions. You should also make sure that you have not used the names for something else earlier in your session.

Mathematica functions can have any number of arguments.

```
In[6]:= hump[x_, xmax_] := (x - xmax)^2 / xmax
```

You can use the `hump` function just as you would any of the built-in functions.

```
In[7]:= 2 + hump[x, 7/(3+5)]

                 7        2
             8 (-(-) + x)
                 8
Out[7]= 2 + --------------
                   7
```

This gives a new definition for `hump`, which overwrites the previous one.

```
In[8]:= hump[x_, xmax_] := (x - xmax)^4
```

The new definition is displayed.

```
In[9]:= ?hump
hump
                                      4
hump/: hump[x_, xmax_] := (x - xmax)
```

This clears all definitions for `hump`.

```
In[9]:= Clear[hump]
```